Slides Mostly Lec 19 + Lec 20

① we saw that Fourier Tx

= Given a time domain signal
Find which complex sinusoids need to be combined to form this signal.

→ FT records the amplitudes & phases of the required complex sinusoids.

② Defn.
→ FT: write $x(t)$ as a combination of complex sinusoids $(e^{j\omega t})$.

$$\Rightarrow \quad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega \qquad \text{IFT}$$

2π appears b/c of $\omega = 2\pi f$

$$\Rightarrow \quad X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \qquad \text{FT}$$

③ → $X(\omega)$ is a complex variable.

$$X(\omega) = Re(X(\omega)) + j\,Im(X(\omega))$$

$$X(\omega) = |X(\omega)|\, e^{j\angle X(\omega)}$$

→ stores information on magnitudes & phases of complex sinusoids required to form the signal.

④ Let's get to know the mathematics of FT !!

ⓐ What is $\int_{-\infty}^{\infty} e^{-j\omega t}\, dt = ?$ —— ★

⇒ There are two cases:

for $\omega \neq 0$ $\quad e^{-j\omega t}$ is a periodic function

$$e^{-j\omega t} = \cos(\omega t) + j\sin(\omega t)$$

for $\omega = 0$ $\quad e^{-j\omega t} = e^{0} = 1$ is a constant.

$$e^{-j(0)t} = \cos(\cancel{0})^{1} + j\sin(\cancel{0})^{0} = 1.$$

⇒ You can convince yourselves that for $\omega \neq 0$, the total area under $e^{-j\omega t}$ equals zero. (e.g. by drawing real & complex parts)

⇒ for $\omega = 0$ we have

$$\int_{-\infty}^{\infty} e^{0}\, dt = \int_{-\infty}^{\infty} 1 . dt = t \Big|_{-\infty}^{\infty} = \infty$$

⇒ so ★ is zero for all values of ω, except $\omega = 0$ where it has infinite amplitude.

⟶ we know such a function : Continuous-time Impulse

$$\delta(\omega)$$

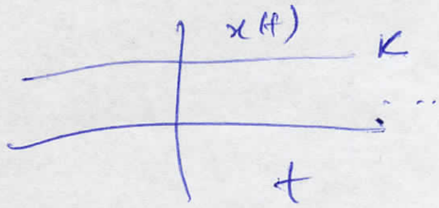
$$\therefore \int_{-\infty}^{\infty} e^{-j\omega t}\, dt = 2\pi\, \delta(\omega)$$

→ the 2π appears due to $\omega = 2\pi f$

→ if we had $\int_{-\infty}^{\infty} e^{-ift}\, dt$ result would be just $\boxed{\delta(\omega)}$

b → using the above, let's find FT of a constant (k)

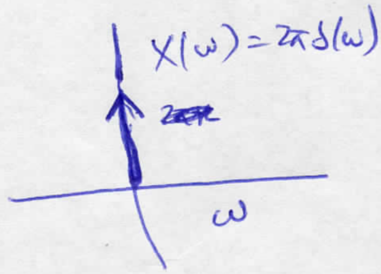
$x(t) = k \quad \forall t$



FT: $X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} k\, e^{-j\omega t}\, dt$

$= k \int_{-\infty}^{\infty} e^{-j\omega t}\, dt$ → same as (*)

$= k\left(2\pi\, \delta(\omega)\right) = 2\pi k\, \delta(\omega) \Rightarrow$

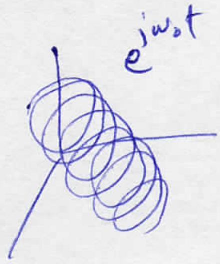


→ this makes sense, since a constant has zero frequency (resulting in $X(\omega)$ having a peak at $\omega = 0$, and zero everywhere else)

(4)

(c) Now, let's find FT of a single complex sinusoid

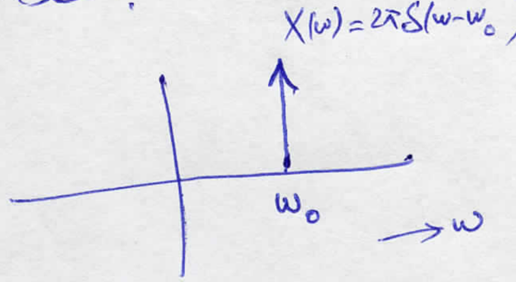
e.g. $x(t) = e^{jw_0 t}$ where w_0 = constant.



$$X(w) = \int_{-\infty}^{\infty} e^{jw_0 t} e^{-jwt} dt = \int_{-\infty}^{\infty} e^{j(w_0 - w)t} dt$$

→ which is similar to (A) except that now $e^{j(w_0 - w)t}$ is a constant at $w = w_0$

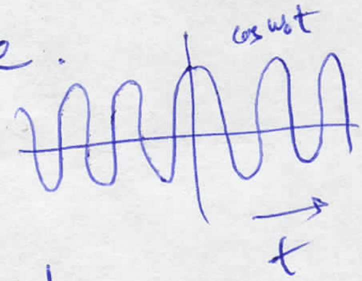
→ This gives a shifted Impulse!



$$X(w) = 2\pi\delta(w - w_0)$$

→ which makes sense, since it says that to form $e^{jw_0 t}$ we just need the complex sinusoid of frequency w_0.

(d) Now let's find FT of a pure cosine.



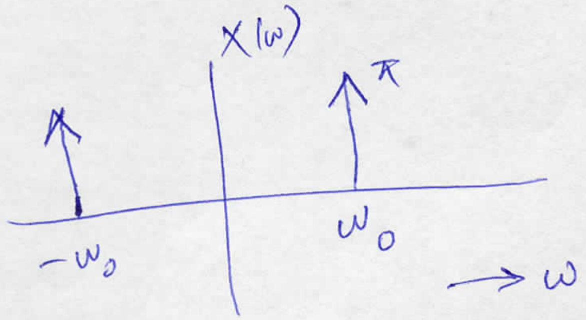
$x(t) = \cos(w_0 t)$ w_0 = constant.

→ Using Euler $\cos(w_0 t) = \frac{1}{2} e^{jw_0 t} + \frac{1}{2} e^{-jw_0 t}$ — (Δ)

→ $X(s) = \int_{-\infty}^{\infty} \cos(w_0 t)\, dt = \frac{1}{2}\int_{-\infty}^{\infty} e^{jw_0 t} dt + \frac{1}{2}\int_{-\infty}^{\infty} e^{-jw_0 t} dt$

using part (c) → $X(s) = \frac{1}{2}\left(2\pi\delta(w + w_0) + 2\pi\delta(w - w_0)\right) = \pi\left(\delta(w + w_0) + \delta(w - w_0)\right)$

→ which make sense since (A) also tells us that to form $\cos(\omega_0 t)$ you need two complex sinusoids, one at ω_0 and one at $-\omega_0$



(C) Let's find FT of an exponential (not complex)

$$x(t) = e^{-at}u(t) \qquad a : \text{Real}$$

→ There are two cases

case (i) $a > 0$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j\omega t}\,dt = \int_{0}^{\infty} e^{-at}e^{-j\omega t}\,dt$$

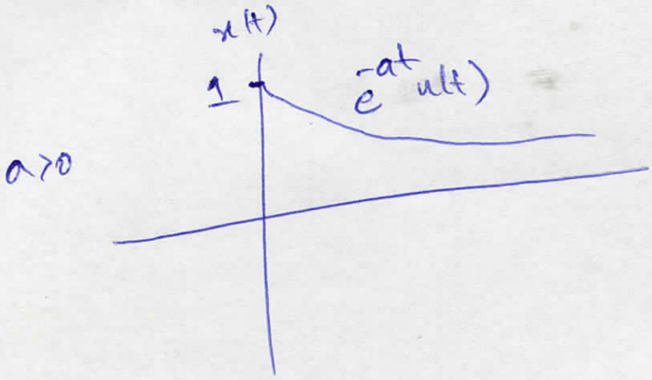
$$= -\frac{1}{a+j\omega}\, e^{-at}e^{-j\omega t}\Big|_{0}^{\infty}$$

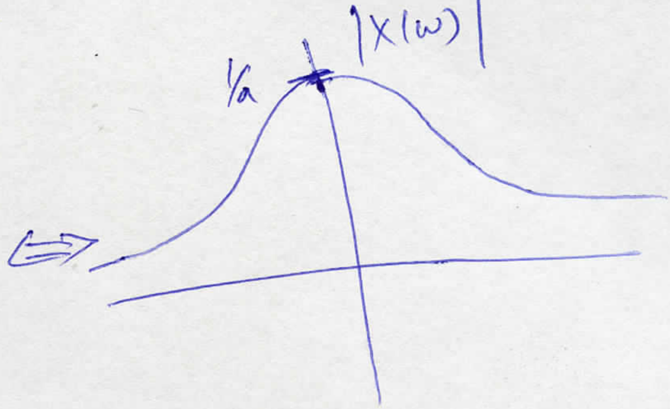
→ oscillating function with modulus always ~~less than~~ equal to 1.

→ exponentially decaying since $a > 0$ and $t \geq 0$

$$\Rightarrow X(\omega) = \frac{1}{a+j\omega} \qquad a > 0$$

$$\Rightarrow \left|X(\omega)\right| = \frac{1}{\sqrt{a^2+\omega^2}}$$





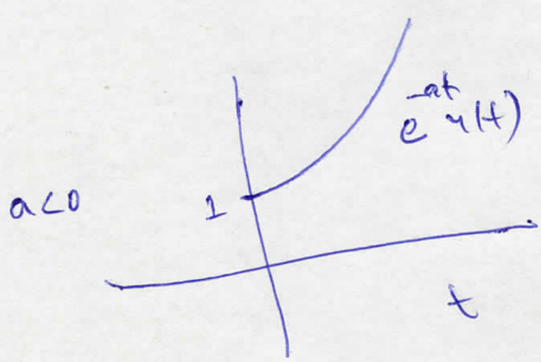
Case (ii) $a < 0$

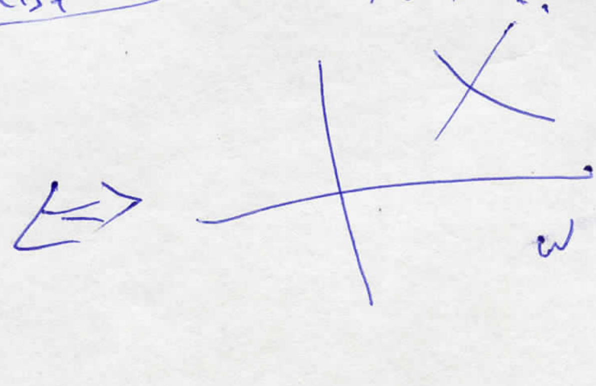
$$X(\omega) = \int_{-\infty}^{\infty} e^{-at}u(t)\, e^{-j\omega t}\, dt = \int_{0}^{\infty} e^{-at} e^{-j\omega t}\, dt$$

$$= -\frac{1}{a+j\omega}\, e^{-at} e^{-j\omega t}\Big|_{0}^{\infty}$$

→ exponentially growing function since $a<0$ and $t \geq 0$

$$\Rightarrow X(\omega) = \infty = \boxed{\text{FT Does Not Exist}}$$



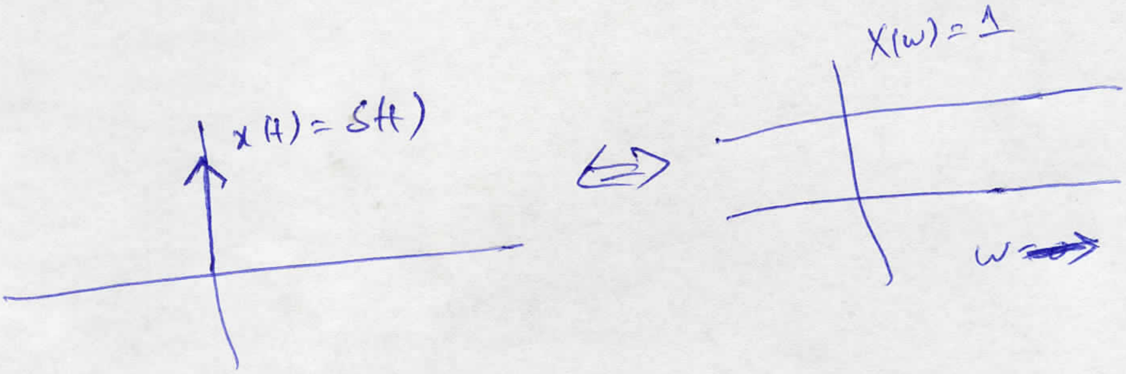


(f) Let's find FT of $\delta(t)$

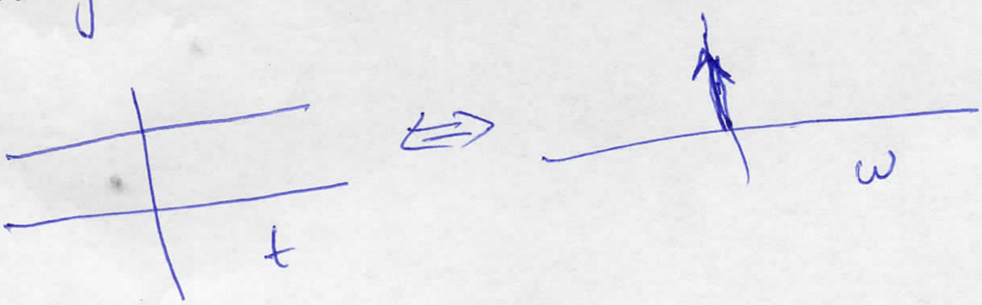
$$x(t) = \delta(t)$$

Sampling property of Impulse!

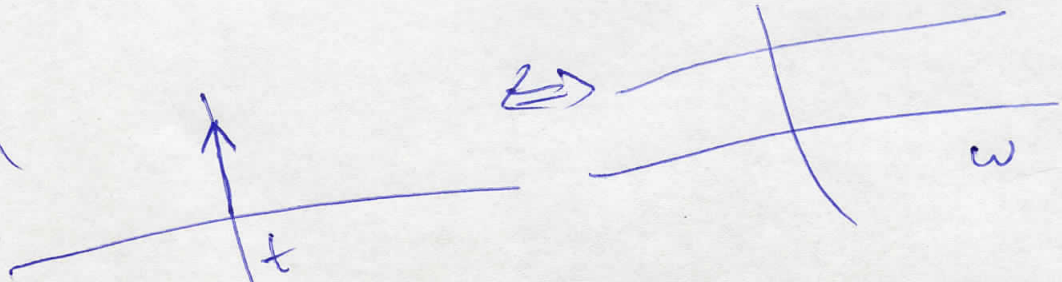
$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega(0)}\, dt = \int_{-\infty}^{\infty} \delta(t)\, dt = 1$$



⇒ Duality note from (b) & (f) that



and



→ In fact this two-way relation (with appropriate scaling by 2π) holds for all FT pairs! ← called <u>Duality</u>

$$x(t) \Longleftrightarrow X(\omega)$$